THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS CHAPTER TEST SET B

Chapter 1: **Orienting Yourself: The Use of Coordinates** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The point $(4, -7)$ lies: 1

(a) Quadrant III **(b)** Quadrant I **(c)** Quadrant IV **(d)** Quadrant II

2. The point $(-5, 0)$ lies: 1

(a) on the x-axis **(b)** Quadrant III

(c) Quadrant II **(d)** Quadrant I

3. The distance between $A(5, 1)$ and $B(-7, 6)$ is: 1

(a) 15 units **(b)** 12 units **(c)** 14 units **(d)** 13 units

4. The midpoint of the segment joining $A(2, 4)$ and $B(6, 10)$ is: 1

(a) $(2, 3)$ **(b)** $(8, 14)$ **(c)** $(4, 7)$ **(d)** $(4, -7)$

5. The midpoint of the segment joining $A(-6, -2)$ and $B(2, 8)$ is: 1

(a) $(-4, 6)$ **(b)** $(-2, -3)$ **(c)** $(-2, 3)$ **(d)** $(4, 5)$

6. The distance of the point $(6, -8)$ from the origin is: 1

(a) 11 units **(b)** 6 units **(c)** 14 units **(d)** 10 units

7. The reflection of the point $(3, -4)$ in the y-axis is: 1

(a) $(-3, -4)$ **(b)** $(-3, 4)$ **(c)** $(3, 4)$ **(d)** $(-4, 3)$

8. The coordinates of the origin are: 1

(a) $(1, 0)$ **(b)** $(1, 1)$ **(c)** $(0, 1)$ **(d)** $(0, 0)$

9. The x-coordinate of a point is its perpendicular distance from the: 1

(a) y-axis (measured along the x-axis)

(b) origin

(c) x-axis

(d) line $y = x$

10. Two vertices of a rectangle are $(1, 2)$ and $(5, 2)$ and one side has length 3 above the x-axis. 1
The other two vertices are:

(a) $(1, -1)$ and $(5, -1)$ **(b)** $(4, 2)$ and $(8, 2)$

(c) $(1, 3)$ and $(5, 3)$ **(d)** $(1, 5)$ and $(5, 5)$

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the coordinates of the midpoint of the segment joining $P(4, 6)$ and $Q(10, -2)$. 2

12. Find the coordinates of the midpoint of the segment joining $P(-5, 3)$ and $Q(1, 9)$. 2

13. Find the coordinates of the midpoint of the segment joining $P(7, -4)$ and $Q(-3, 2)$. 2

14. Find the coordinates of the midpoint of the segment joining $P(0, 0)$ and $Q(-8, 10)$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Show that $A(-2, -1)$, $B(3, -1)$, $C(3, 2)$, $D(-2, 2)$ are the vertices of a rectangle. 3

16. Show that $A(0, 2)$, $B(6, 2)$, $C(6, 7)$, $D(0, 7)$ are the vertices of a rectangle. 3

17. Find the coordinates of the points of trisection of the segment joining $A(2, 5)$ and $B(11, 14)$. 3

OR

Find the coordinates of the points of trisection of the segment joining $A(-4, 1)$ and $B(8, 7)$. 3

18. The midpoints of the sides of a triangle are $D(5, 1)$, $E(6, 5)$ and $F(0, 3)$. Find the vertices. 3

OR

Show that the points $M(-3, -4)$, $A(0, 0)$ and $G(6, 8)$ are collinear. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Show that the points $A(-3, -4)$, $B(0, 0)$, $C(3, 4)$ lie on a straight line, and find the ratio in which B divides AC. 5

OR

The vertices of a triangle are $A(0, 0)$, $B(8, 0)$, $C(4, 6)$. 5

(i) Find the lengths of the sides and state its type.

(ii) Find the midpoints D, E, F of BC, CA, AB and the perimeter of triangle DEF.

20. Find the coordinates of the centre of the circle passing through $A(0, 0)$, $B(6, 0)$ and $C(0, 8)$ and its radius. 5

OR

Plot (describe) the points $A(-3, 2)$, $B(1, 2)$, $C(1, -4)$, $D(-3, -4)$. 5

(i) Name the figure.

(ii) Find its area and the length of the diagonal.

(iii) Write the coordinates of its centre.

END OF QUESTION PAPER — Total: 40 Marks